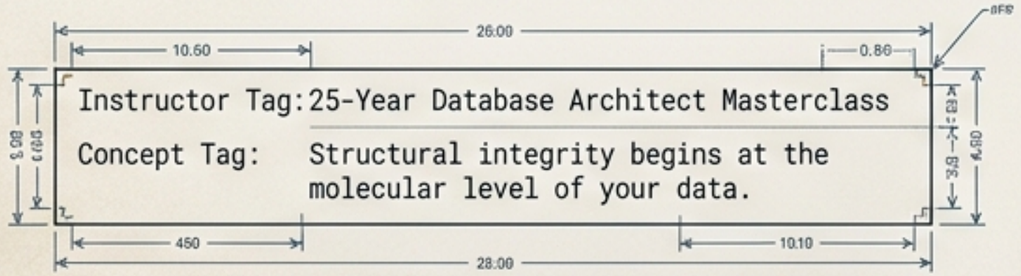
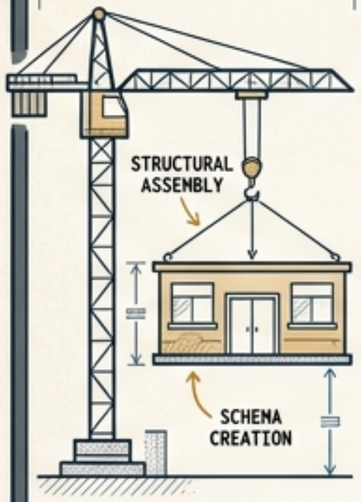
Architecture of Precision

SQL Command Hierarchies & MySQL Data Types


26.00
10.60
0.80
Instructor Tag: 25-Year Database Architect Masterclass
Concept Tag: Structural integrity begins at the molecular level of your data.
4.60
3.8
4.50
3.80
4.50
4.60
4.50
4.50
4.50

Teacher's Brain Snapshot: SQL is not a single tool—it is a toolkit divided into five functional command families.

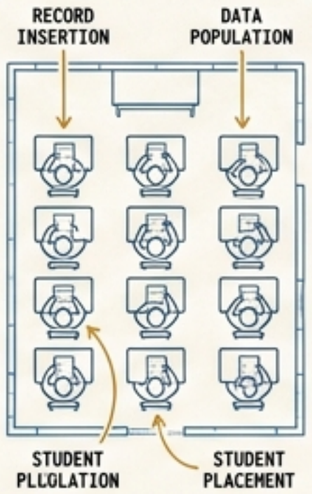


DDL
(Data Definition Language)

Acts directly on the database schema. Creates, alters, or destroys empty structural buckets.

TECHNICAL SPECIFICATION: CODE

CREATE, ALTER, DROP



DML
(Data Manipulation Language)

Norks with individual records within structures. Enrolling and seating the students.

TECHNICAL SPECIFICATION: CODE

INSERT, UPDATE, DELETE



DQL
(Data Query Language)

The search engine used to find and filter data without changing it. Calling attendance.

TECHNICAL SPECIFICATION: CODE

SELECT

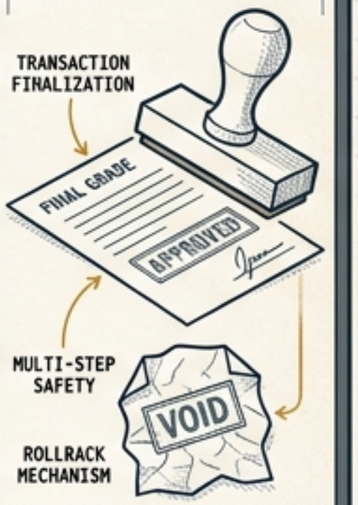


DCL
(Data Control Language)

Adainisters system security and file perxissions. Locks and staff keycards.

TECHNICAL SPECIFICATION: CODE

GRANT, REVOKE



TCL
(Transaction Control Language)

Manages transactions to ensure multi-step safety. Stonping final grades or crumpling the paper.

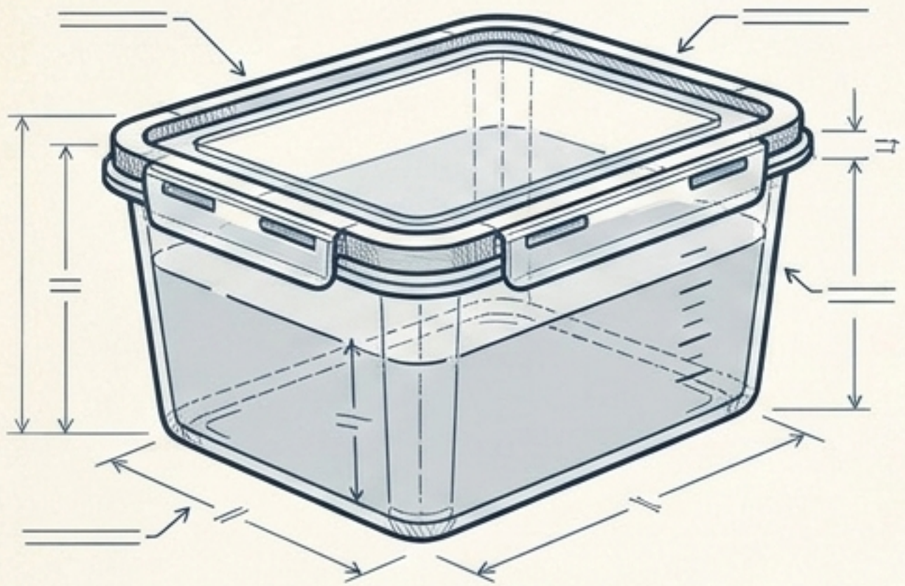
TECHNICAL SPECIFICATION: CODE

COMMIT, ROLLBACK

Gemini Notebook

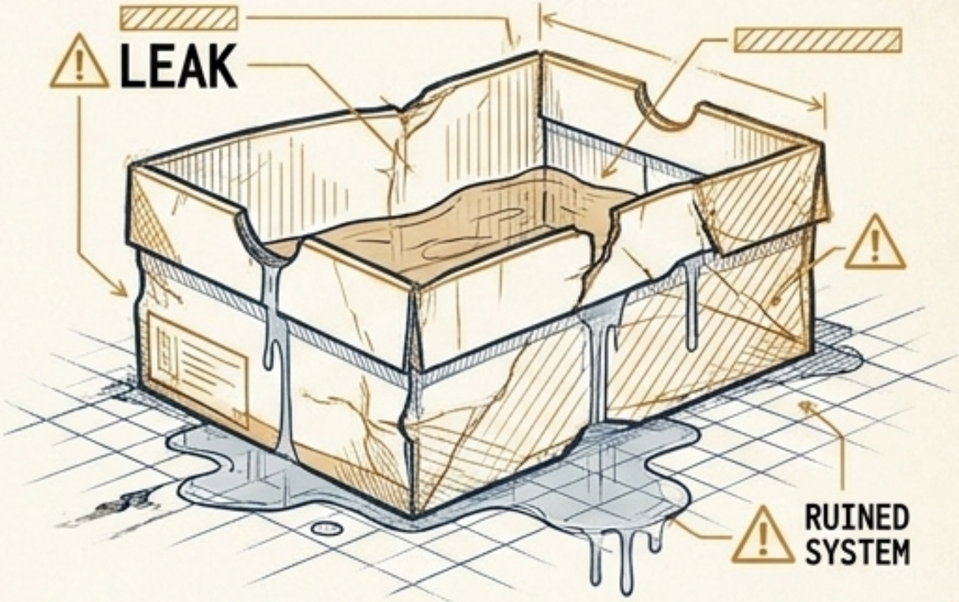
Teacher's Brain Snapshot: A database is a collection of strict Tupperware containers; you cannot put hot soup into a cardboard shoe box.

STRICT KITCHEN STORAGE



Unlike spreadsheets, databases require rigid schema-definition. A Data Type dictates to the DBMS exactly how much storage space to allocate on the physical drive, what processing rules apply, and how to validate the inputs.

CATASTROPHIC FAILURE



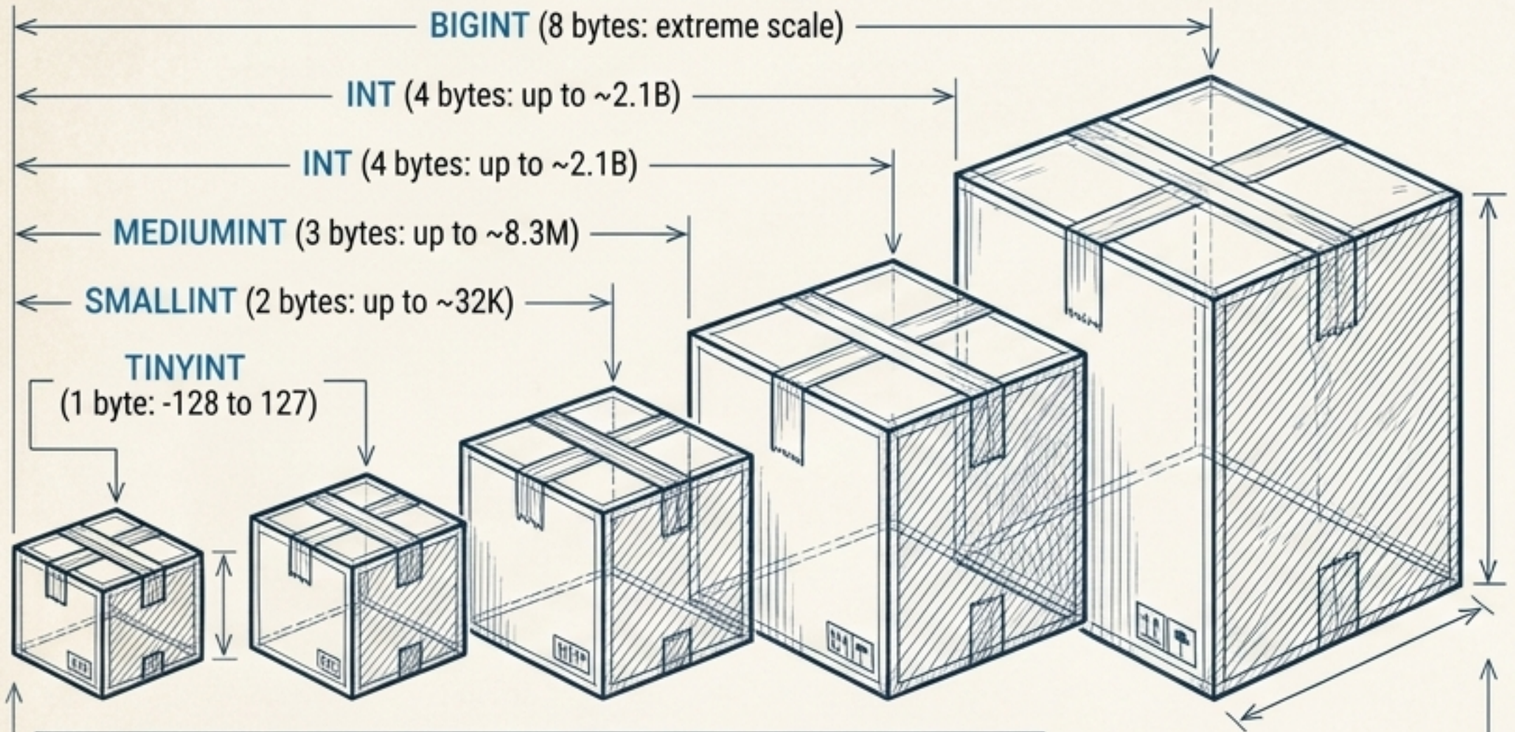
If you put soup in a cardboard shoe box, it will leak and ruin the system. Data types ensure the physical hard disk is optimized so every piece of data has its perfect, watertight container.

TECHNICAL SPECIFICATION: CODE

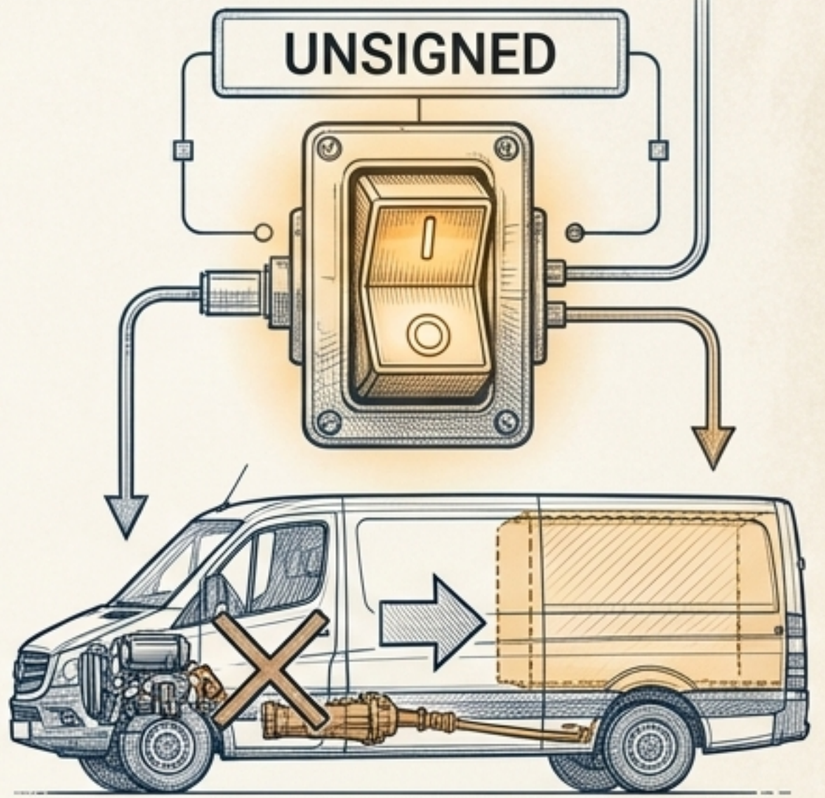
```
CREATE TABLE Users ( UserID INT UNSIGNED PRIMARY KEY );
```

Gemini Notebook

TEACHER'S BRAIN SNAPSHOT: CHOOSE THE SMALLEST INTEGER CONTAINER THAT SAFELY HOLDS YOUR DATA TO PREVENT CATASTROPHIC MEMORY BLOAT.



TECHNICAL SPECIFICATION: INTEGER CONTAINERS



TECHNICAL SPECIFICATION: CODE

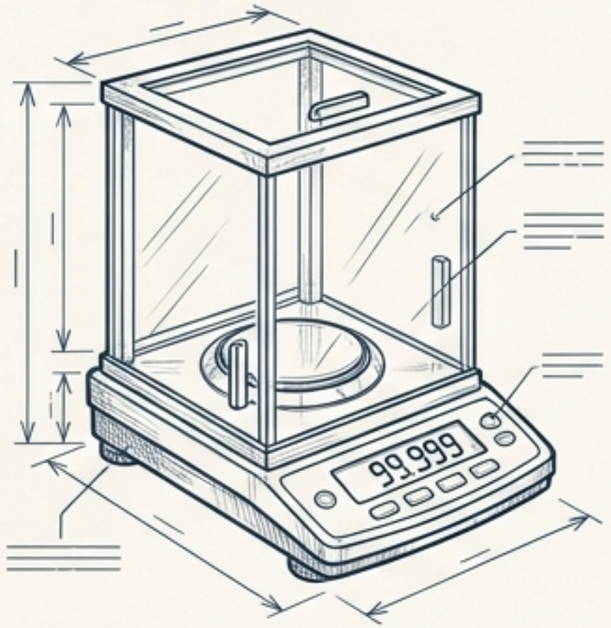
```
CREATE TABLE Inventory (
    ItemID MEDIUMINT UNSIGNED PRIMARY KEY,
    Quantity TINYINT UNSIGNED NOT NULL,
    TotalViews BIGINT DEFAULT 0
);
```

CLASSROOM METAPHOR: MATRYOSHKA SHIPPING BOXES. USING UNSIGNED REMOVES THE NEGATIVE RANGE. IT'S LIKE REMOVING THE REVERSE GEAR FROM A DELIVERY VAN—YOU CAN'T DRIVE BACKWARDS, BUT YOU INSTANTLY DOUBLE YOUR NON-NEGATIVE CARGO SPACE.

Gemini Notebook

TEACHER'S BRAIN SNAPSHOT:
NEVER USE A BATHROOM SCALE TO WEIGH DIAMONDS; NEVER USE FLOAT FOR MONEY.

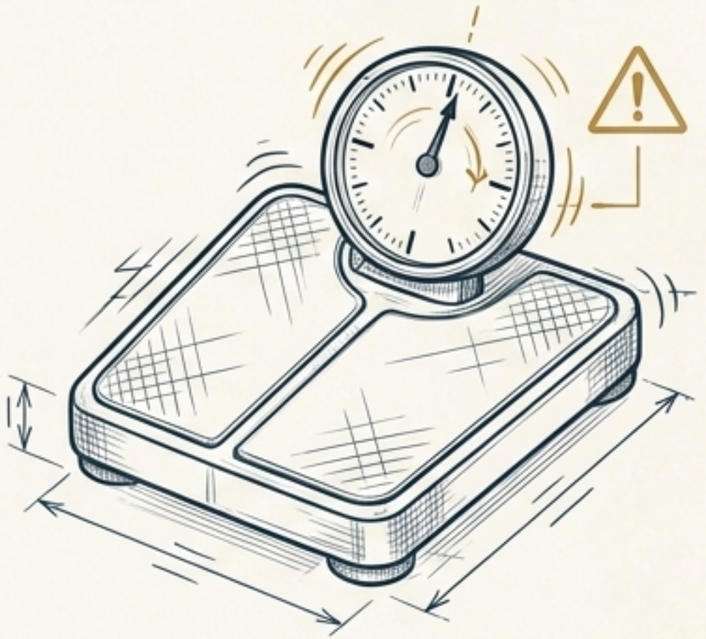
EXACT PRECISION



DECIMAL / NUMERIC

Stores exact numeric precision. Essential for financial and strictly accurate fractional data. 10.05 is recorded exactly as 10.05.

APPROXIMATE SCALE



FLOAT / DOUBLE

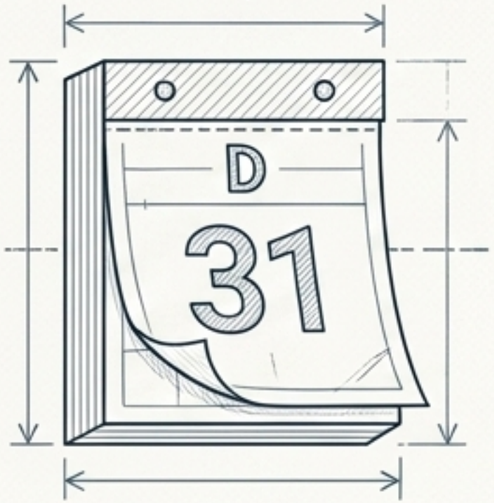
Stores approximate numeric values using floating-point math. Trades absolute exactness for vast mathematical scale and processing speed. 10.05 might be saved as 10.04999999 behind the scenes.

TECHNICAL SPECIFICATION: CODE

```
CREATE TABLE Financials (
    AccountID INT PRIMARY KEY,
    AccountBalance DECIMAL(15, 2) NOT NULL,
    ScientificMeasurement DOUBLE
);
```

Gemini Notebook

TEACHER'S BRAIN SNAPSHOT:
DATETIME IS A PHOTOGRAPH OF A CLOCK;
TIMESTAMP IS A LIVE TIMEZONE TRANSLATOR.



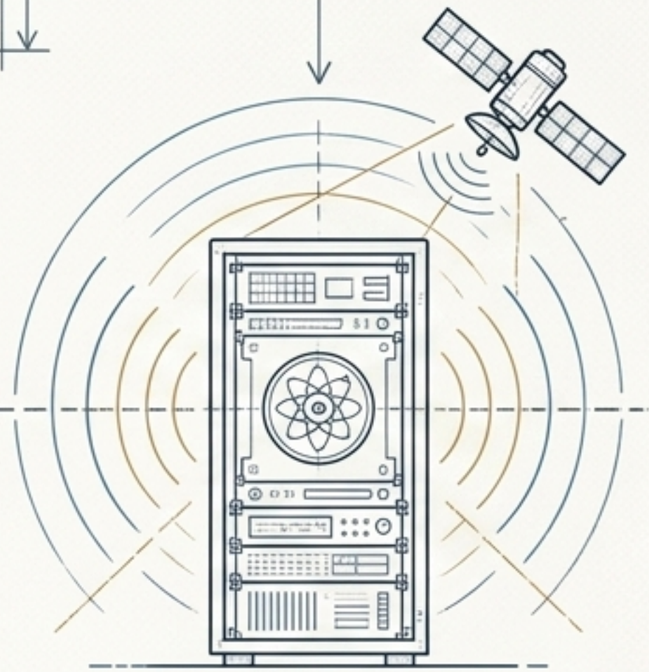
DATE

Stores specific temporal fragments (YYYY-MM-DD).



DATETIME

Stores a static date and time value. Like a printed concert ticket, it says 9:00 AM no matter where in the world you take it.



TIMESTAMP

Stores date and time, automatically converts to UTC for storage, and back to current timezone for retrieval. If Tokyo writes data, New York reads it accurately.

TECHNICAL SPECIFICATION: CODE

```
CREATE TABLE UserSessions (
    SessionID INT PRIMARY KEY,
    BirthDate DATE,
    StaticMeetingTime DATETIME,
    LastModified TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
);
```

Gemini Notebook

TEACHER'S BRAIN SNAPSHOT:

CHAR charges you for the whole room; VARCHAR only charges you for the space you actually take up.

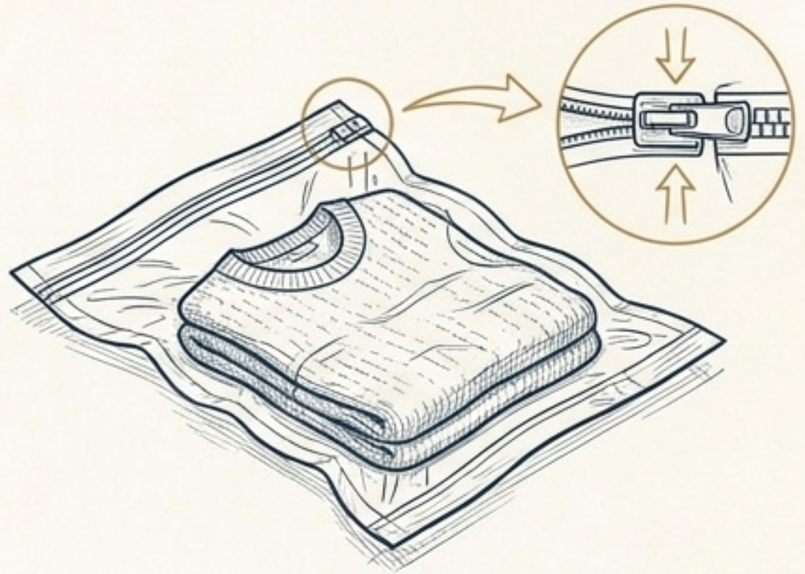
FIXED-LENGTH ALLOCATION



CHAR

If you define CHAR(50) and insert 'Hi', the database pads the remaining 48 spaces on the physical disk. Optimized for exact-length data like hashes or state codes.

VARIABLE-LENGTH ALLOCATION



VARCHAR

Stores only the characters provided plus a tiny 1-2 byte prefix recording the length (the zipper). Optimizes disk space for dynamic, varying text.

TECHNICAL SPECIFICATION: CODE

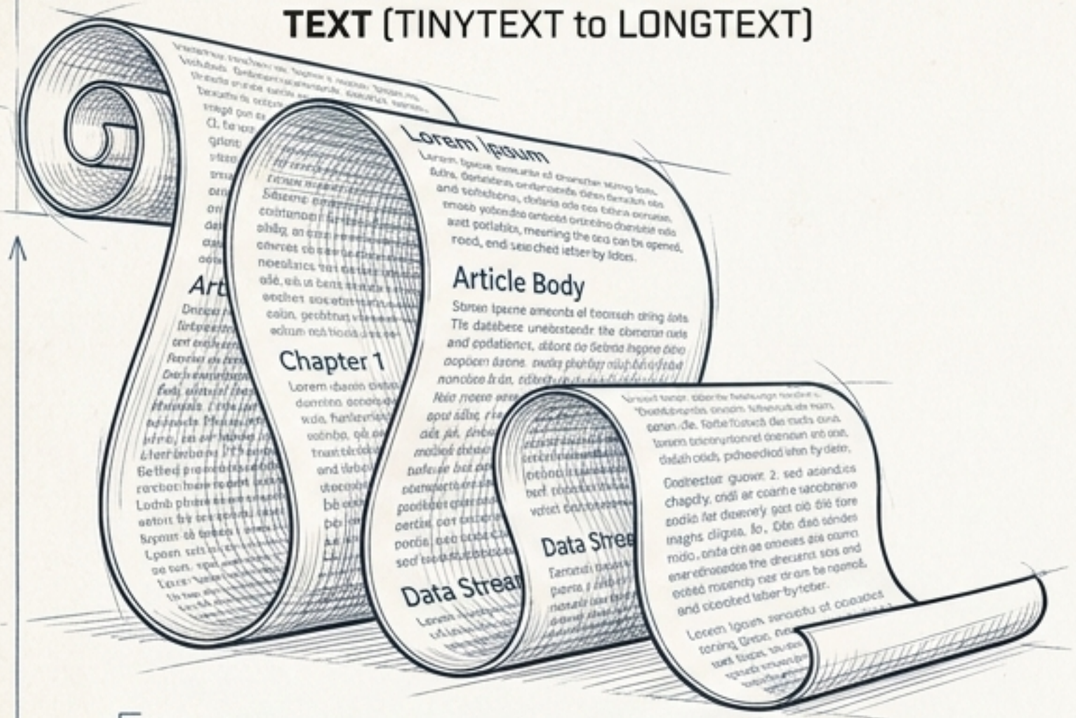
```
CREATE TABLE Passports (
    CountryCode CHAR(2) NOT NULL,
    CitizenName VARCHAR(100) NOT NULL
);
```

Gemini Notebook

TEACHER'S BRAIN SNAPSHOT: TEXT is for encyclopedias you can read; BLOB is for raw files you can only decode.

CHARACTER DATA

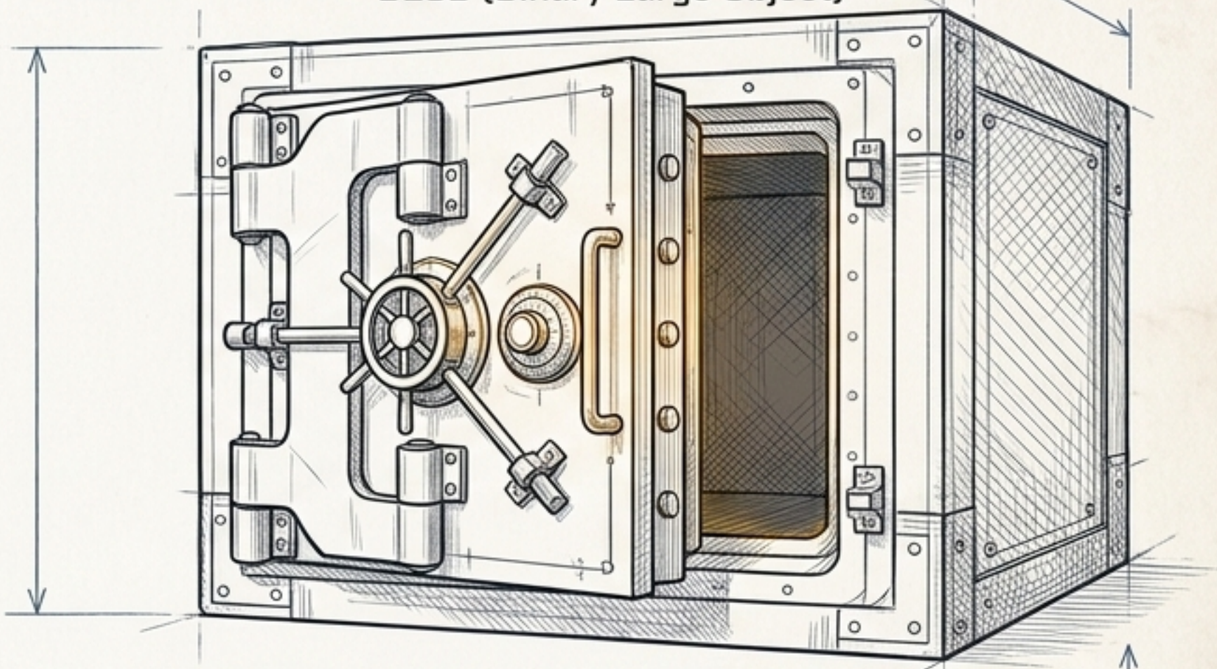
TEXT (TINYTEXT to LONGTEXT)



Stores massive amounts of character string data. The database understands the character sets and collations, meaning the text can be opened, read, and searched letter-by-letter.

BINARY DATA

BLOB (Binary Large Object)



Stores raw binary strings (images, PDFs, compiled code). Possesses no character set and relies strictly on numeric byte values. The database just holds the heavy weight until an application extracts it.

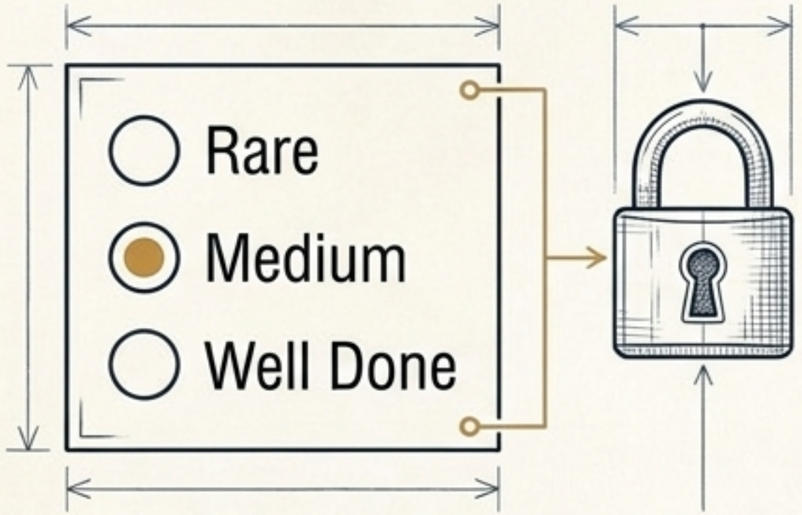
TECHNICAL SPECIFICATION: CODE

```
CREATE TABLE BlogPosts (
  PostID INT    PRIMARY KEY,
  ArticleBody   MEDIUMTEXT,
  HeroImageRaw  BLOB
);
```

Gemini Notebook

TEACHER'S BRAIN SNAPSHOT: ENUM is a strict multiple-choice test; SET is a multi-select buffet.

EXCLUSIVE CONSTRAINT



ENUM (Choose One)

Restricts a column to exactly ONE value from a predefined string list. Like ordering steak temperature, it forces exclusivity—it cannot be both rare and well-done.

INCLUSIVE COMBINATIONS



SET (Choose Any)

Allows a column to contain zero, one, or multiple values from a predefined list. Like a pizza topping buffet, you can combine any permitted values.

TECHNICAL SPECIFICATION: CODE

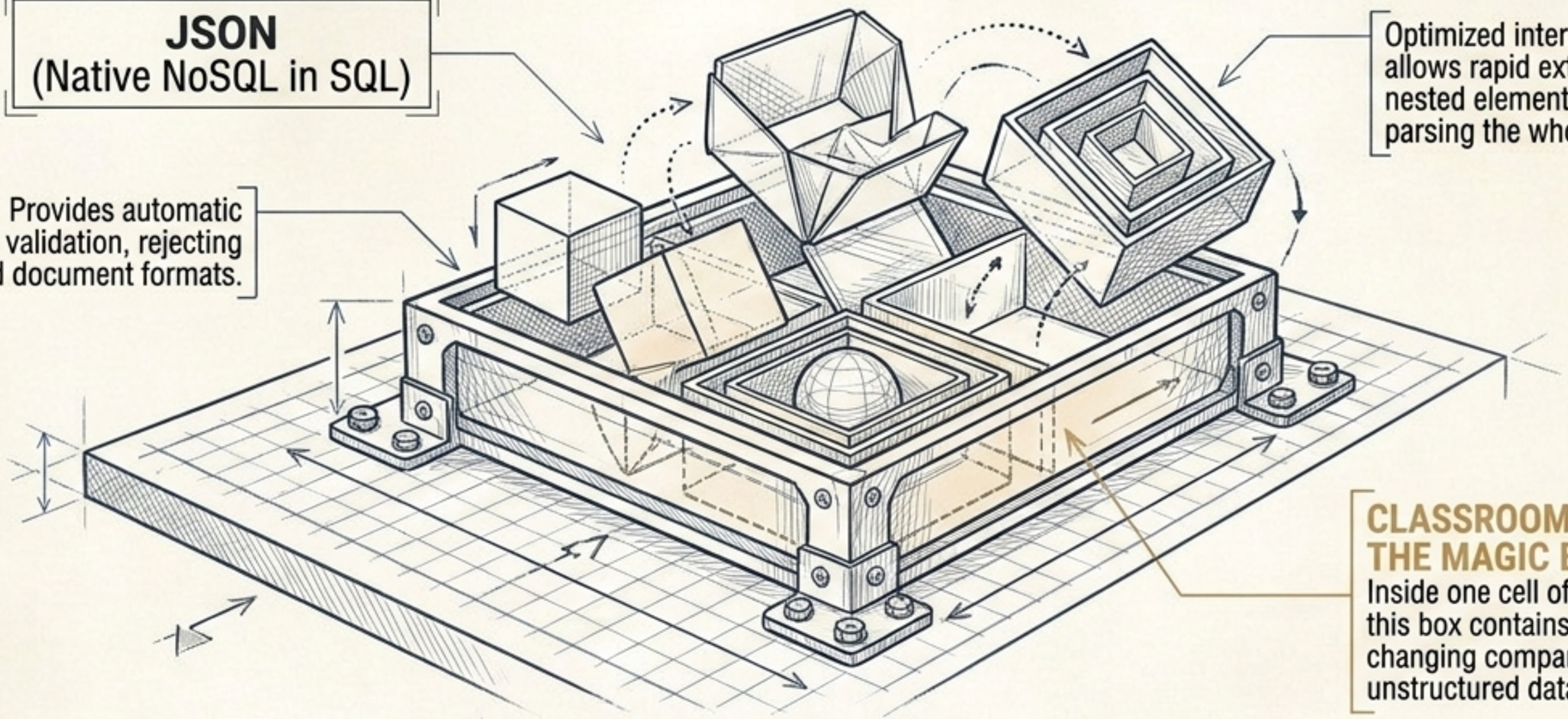
```
CREATE TABLE CustomerPreferences (
   CustomerID INT PRIMARY KEY,
   ShirtSize ENUM('Small', 'Nedium', 'Large') NOT NULL,
   Allergies SET('Peanuts', 'Dairy', 'Gluten', 'Shellfish')
);
```

Gemini Notebook

TEACHER'S BRAIN SNAPSHOT: JSON is a shape-shifting box that lets you smuggle NoSQL flexibility into a rigid relational database.

JSON
(Native NoSQL in SQL)

Provides automatic validation, rejecting invalid document formats.



Optimized internal binary format allows rapid extraction of deeply nested elements without parsing the whole text string.

CLASSROOM METAPHOR: THE MAGIC BENTO BOX.
Inside one cell of a rigid database grid, this box contains infinite, dynamically changing compartments of unstructured data.

TECHNICAL SPECIFICATION: CODE

```
CREATE TABLE GameProfiles (
   PlayerID INT PRINARY KEY,
   Username VARCHAR(50),
   DynamicAttributes JSON
);
```

Gemini Notebook

DQL

DML

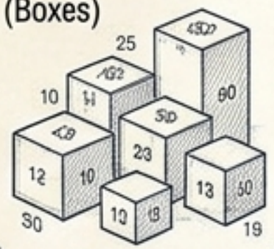
DDL

DCL

TCL

1 Numeric

Integers
(Boxes)

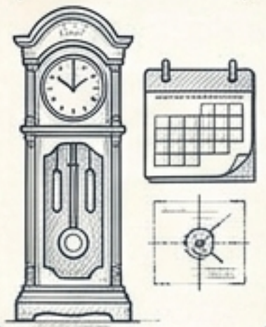


Fixed/Float
(Scales)



2 Temporal

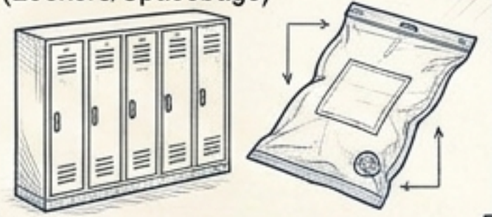
Time
(Clocks/Calendars)



2.75

3 Strings

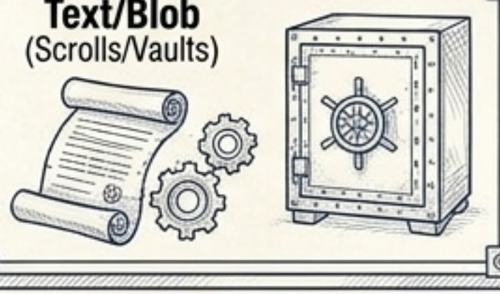
Char/Varchar
(Lockers/Spacebags)



0.75

4 Large Objects

Text/Blob
(Scrolls/Vaults)

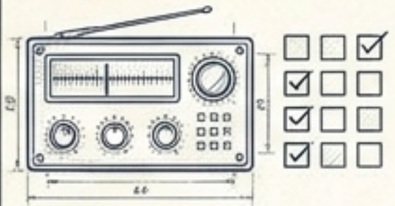


4.16

2.77

5 Special Lists

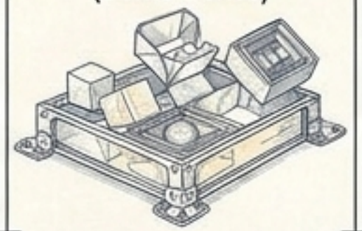
Enum/Set
(Radio/Checkboxes)



2.57

6 Nested

JSON
(Bento Boxes)



2.7

THE COMPLETE ARCHITECTURE.

You now possess the blueprint. You understand the commands that govern the SQL universe. And you know that when you execute a DDL command, the materials you choose dictate the survival, speed, and integrity of your system.

YOU AREN'T JUST WRITING CODE.
YOU ARE ARCHITECTING REALITY.

Gemini Notebook